ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) Resolution of the ADC = $V_{range}/2^n$ = 5 V / 4096 = 1.22×10^{-3} V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity chek: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Sanity check: a 160 mm field of view across 1280 pixels is plausible for a 400 mm standoff and a 16 mm lens.

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\text{ s}^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s; Tustin: $s = (2/T)(z - 1)/(z + 1)$; $a = \omega_c T/2 = 0.0628$.

y_k = **$0.0591\,(u_k + u_{k-1}) + 0.8818\,y_{k-1}$**.

DC gain is unity because it is a low-pass filter.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$; characteristic polynomial $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$L = [18, 82]^T$.

(d) An observer works only when there is no noise, so it has no advantage; a filter is always better because it is simpler.

The filter is safer when the model is perfectly known.

Note: observer poles are usually placed several times faster than the controller poles.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$ so the poles are $s = -2 \pm 2j$.

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 =$ **0.8**.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 =$ **20%**. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Substituting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ **$k_2 = 2$**; $2 + k_1 = 24 \Rightarrow$ **$k_1 = 22$**; $K_I =$ **32**.

(d) In steady state $\dot{\xi} = 0$, so $r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$ for any b and c as long as the system is stable.

Failure mode: integral action amplifies noise and makes the system oscillate. Remedy: put a low-pass filter on the measurement.